

Nuclear Structure Investigations of Light Nuclei with the R3B Experiment

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R³B Experiment @FAIR

Total Interaction Cross Section $^{12}\text{C} + ^{12}\text{C}$

Quasi-Free Scattering $^{12}\text{C}(p,2p)^{11}\text{B}$

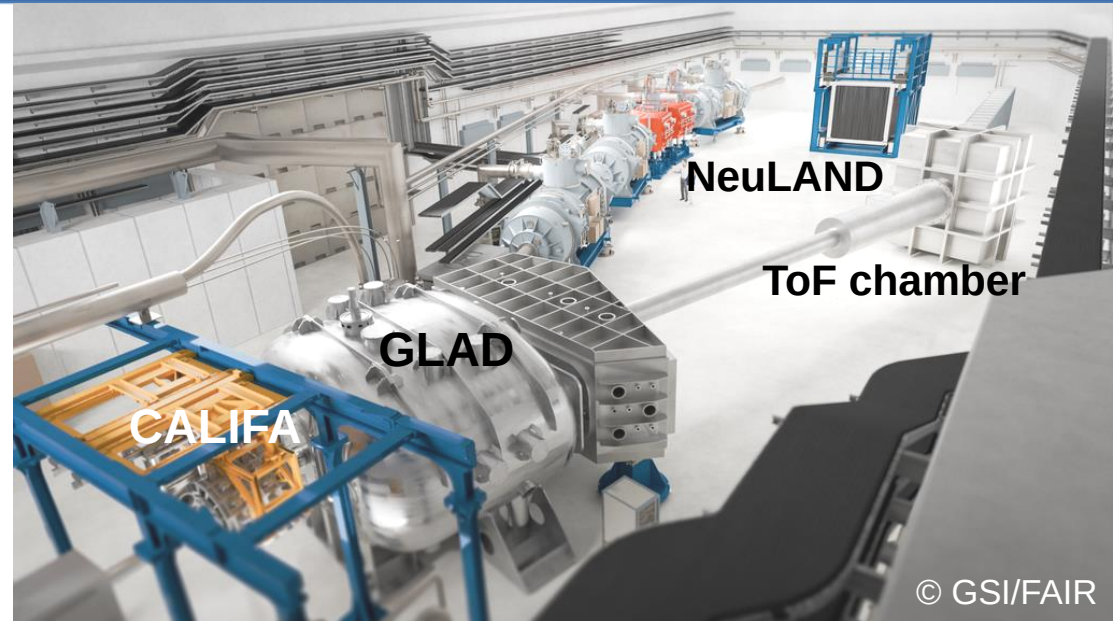
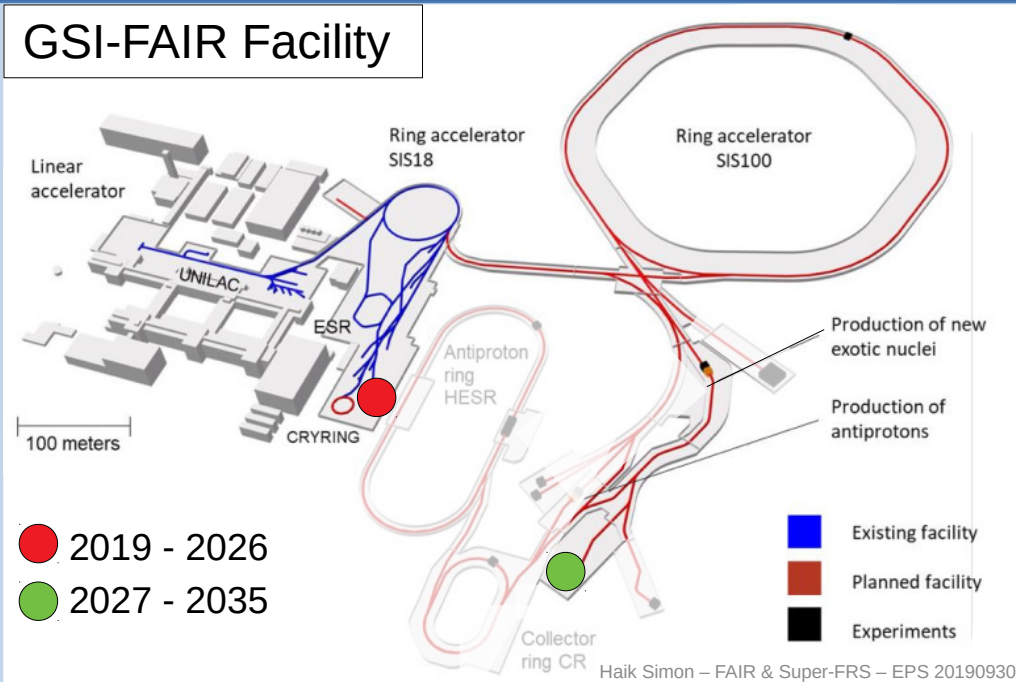
Fission Experiments @ R3B

ML for Cluster Reconstruction in CALIFA

Funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation)
under Germany's Excellence Strategy – EXC-2094 – 390783311,
BMBF 05P19WOFN1, 05P21WOFN1 / BMBF 05P24WO2
and the FAIR Phase-0 program

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GSI-FAIR Facility

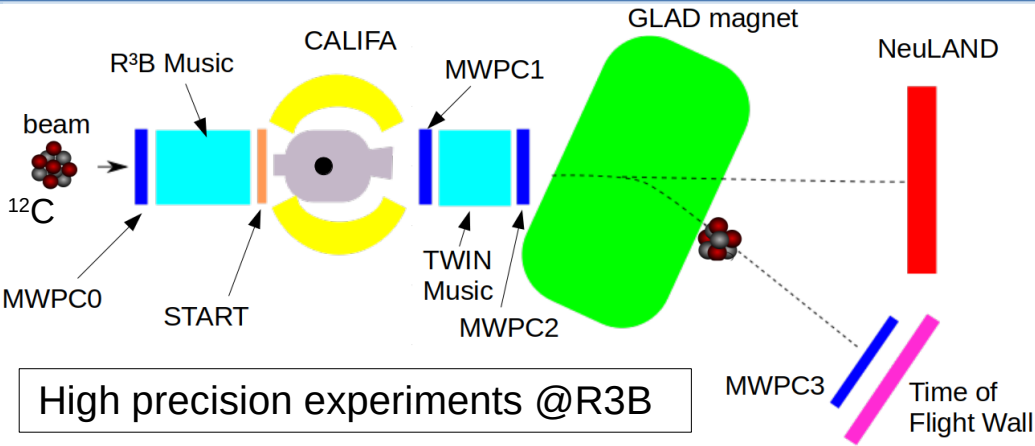


Experimental Approach:

- Inverse kinematics
- Kinematically complete measurements

Scientific goals:

- Study the structure of exotic, unstable nuclei
- Investigate nuclear fission dynamics and fission product properties
- Examine short-range correlations between nucleons



Motivation:

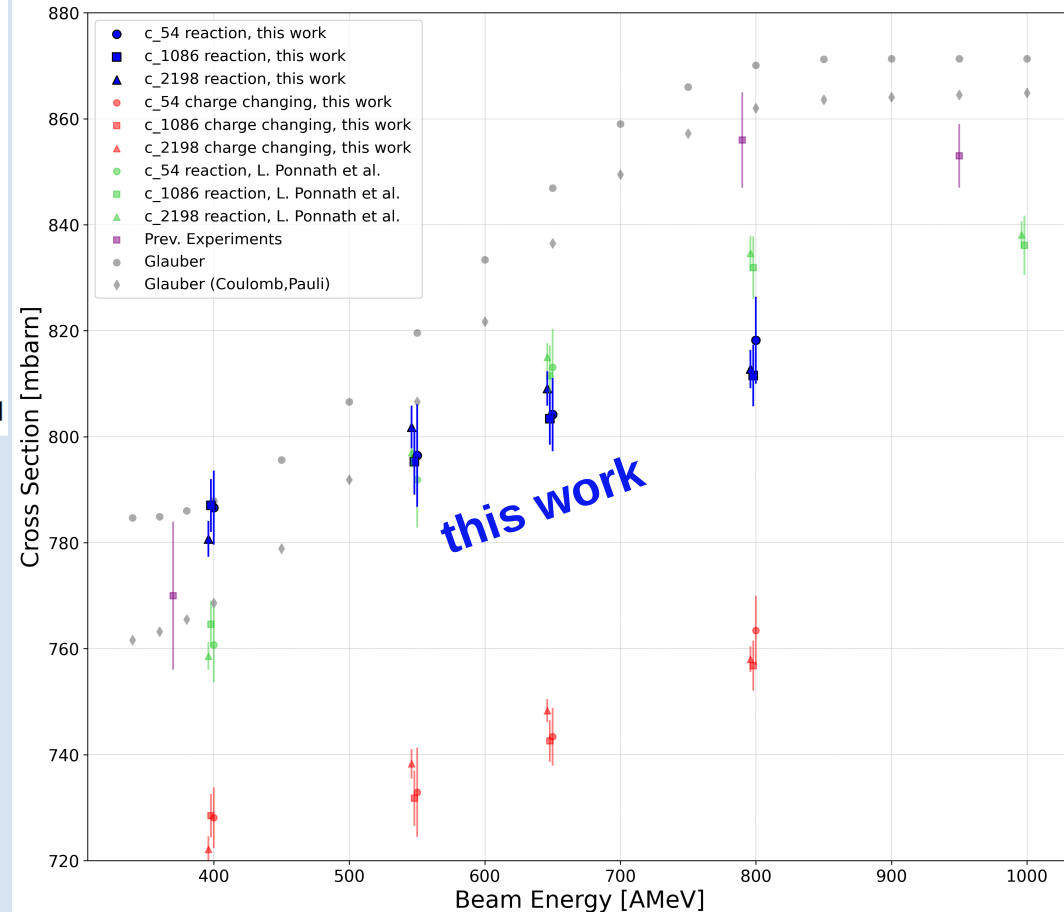
Quantify the uncertainty of the reaction model

Scientific goal:

Constrain slope parameter L of the symmetry energy of the EoS via precise $\sigma_{\Delta N}$ measurements

Measurement via transmission method:

$$N_2 = N_1 e^{-N_t \sigma}$$

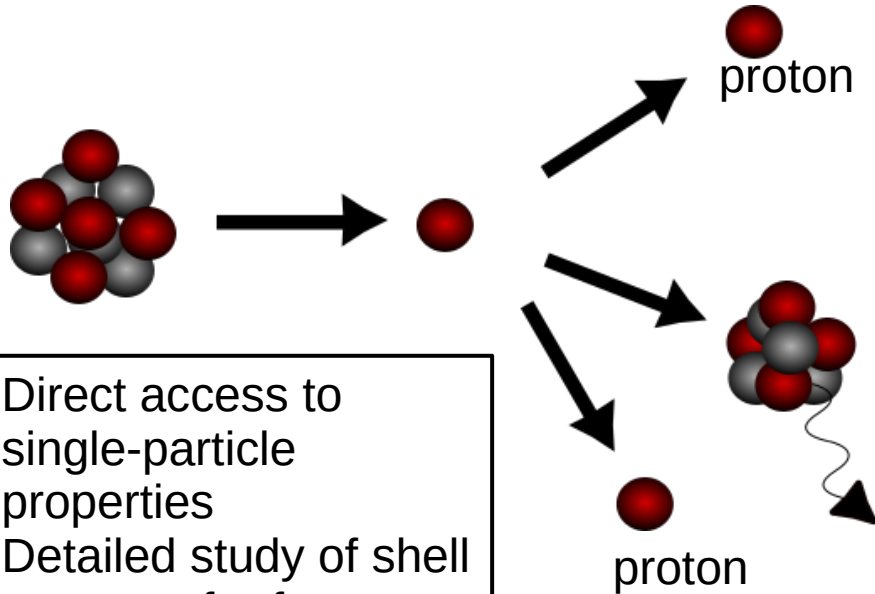


Quasi-Free Scattering $^{12}\text{C}(p,2p)^{11}\text{B}$

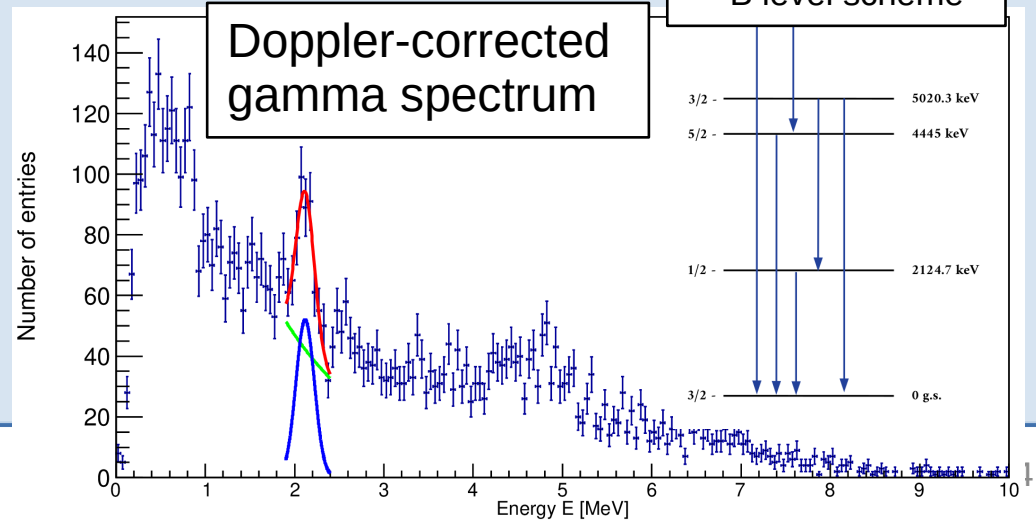
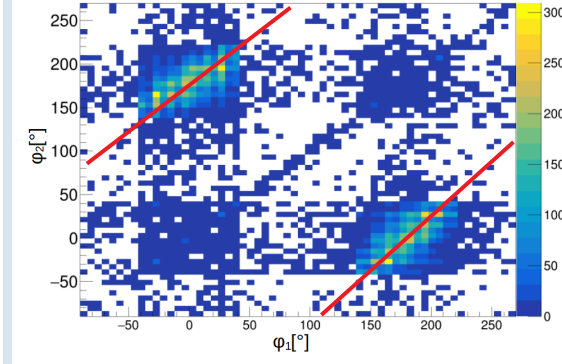
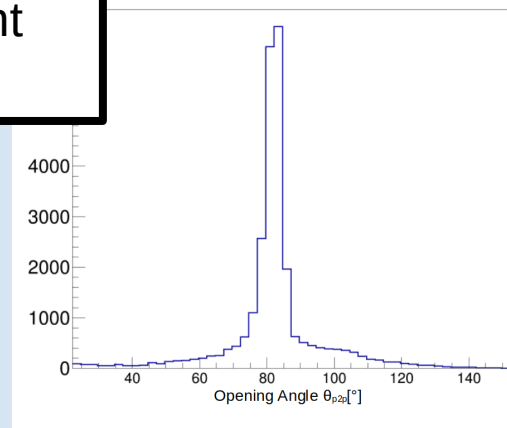
➤ ^{12}C beam
➤ proton like target

➤ 2 protons
➤ ^{11}B fragment (spectator)

two-proton correlations

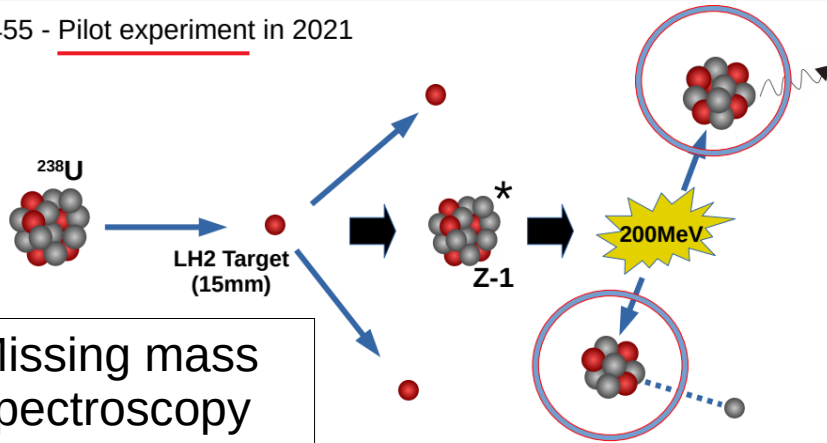


- Direct access to single-particle properties
- Detailed study of shell structure far from stability



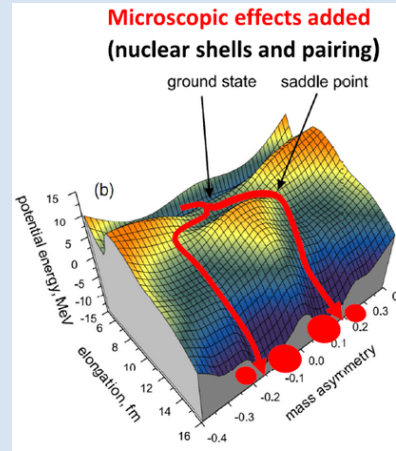
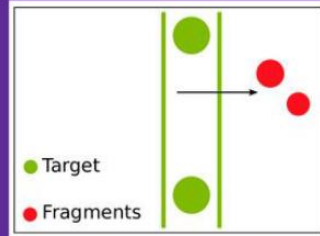
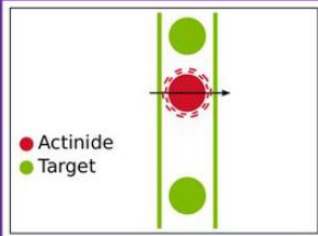
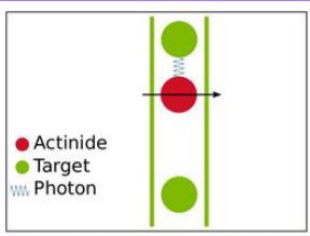
Fission via QFS: $^{238}\text{U}(p, 2pf)$

S455 - Pilot experiment in 2021

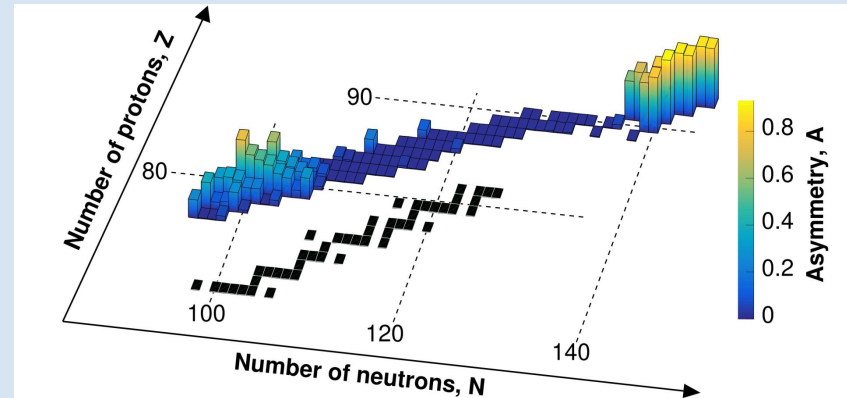
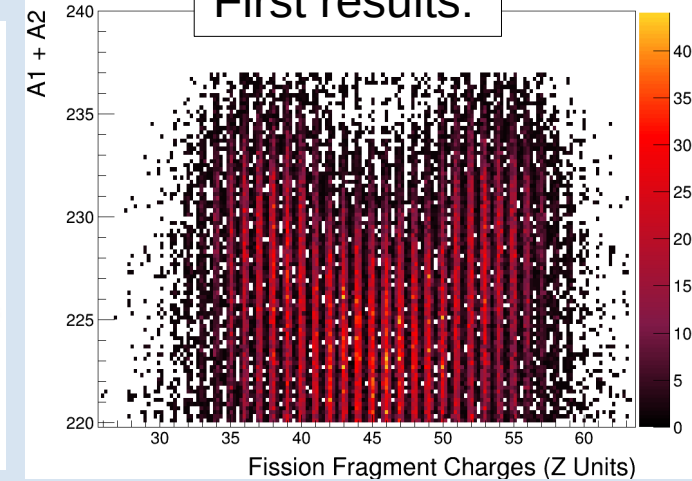


Missing mass spectroscopy

Complementary method via coulomb excitation

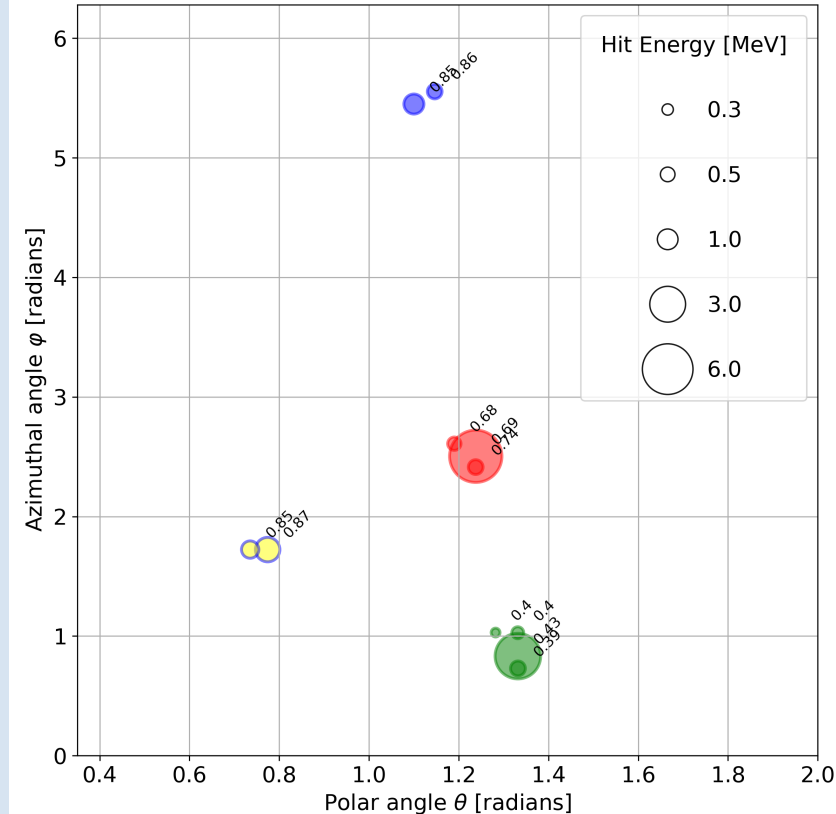
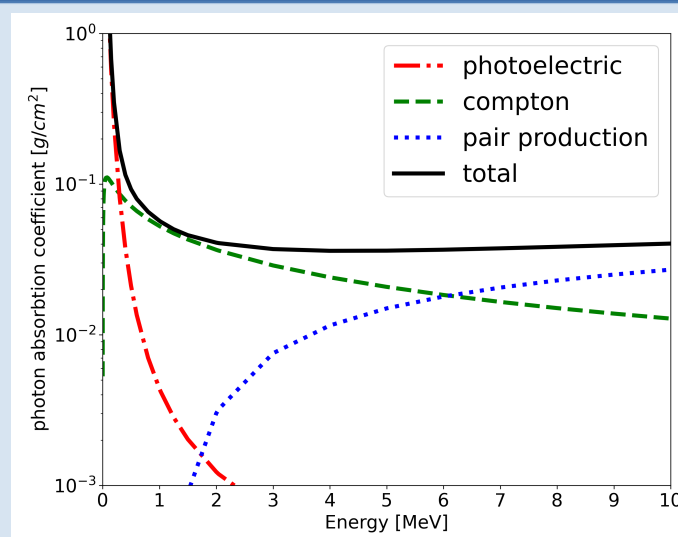
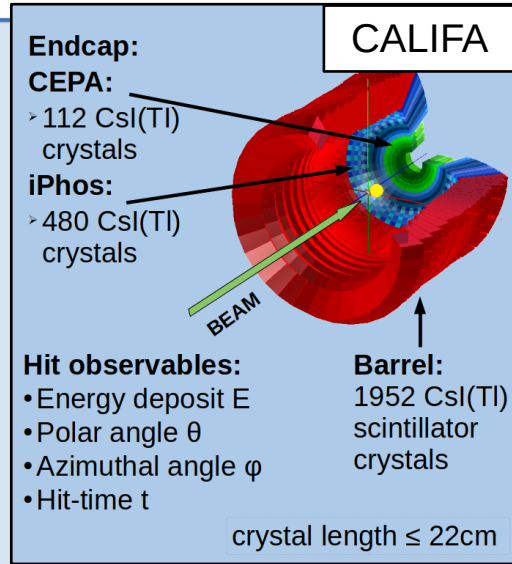


First results:



"An asymmetric fission island driven by shell effects in light fragments", Nature (2025)

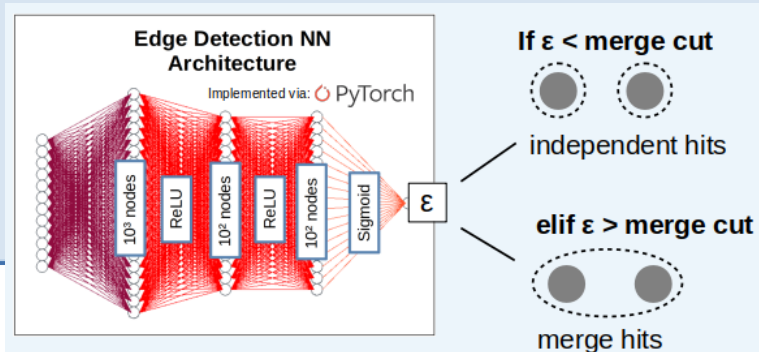
Machine Learning for the Cluster Reconstruction in the CALIFA Calorimeter at R3B



Edge Detection Neural Network

2. Pairwise hit comparison (i, j)

1. Pre-clustering



With support from Origins
Data Science Lab